A National Research Platform for AI Research & Education (NRP AIRE)

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Vision of NRP
Connect every US-accredited institution of higher learning into a world-class and secure Research & Education infrastructure for the entire nation, both rural and urban, across all 50 states. An infrastructure to empower all institutions of higher learning to provide hands-on AI education, and research, providing world-leading access to all students.

In this white paper we show how the Federal Government with its AI Action Plan can fund a necessary transformative change for college education nationwide that can be sustained long term without ongoing Federal funding.

The Challenge the NRP Solves
Artificial Intelligence is primarily an experimental science [1] requiring expensive data and compute infrastructure to equip class labs. The fast and revolutionary adoption of AI across all STEM fields means compute and data access are an essential requirement for modern STEM education. ***Unless we take bold action, AI education, and to a lesser extend STEM education in general, will become a privilege of the elite because only elite campuses can afford to build and support the cyberinfrastructure to do so.***

Based on a survey [2] designed by San Diego State University's David Goldberg that received ~8,000 student responses, we project there is a near-term need to provide formal AI education for up to half of the students enrolled in colleges across the USA. According to the Department of Education, there are almost 3,900 accredited institutions of higher learning in the USA, of which less than 200 are "research-heavy" so-called R1:Doctoral Universities. There are about 20 million students enrolled in these 3,900 institutions. Approximately half of them are enrolled in about a thousand community colleges, providing training across a wide range of professions, as well as offering an affordable entry path to a bachelor's degree program via transfers to 4-year colleges and universities. For example, about half of the incoming students at San Diego State University, and one third at UC San Diego, are transfer students from mostly local community colleges. Motivating and enabling programs that provide a smooth transition between these 2-year and 4-year institutions is a desired outcome of our vision.

However, our vision faces a major challenge: many of the colleges in the US, especially in rural communities lack the expert staff to procure, deploy, and securely operate the kind of advanced computing infrastructure necessary for AI education. Nor do these institutions have the staff to train their educators how to use such infrastructure and incorporate it into their curricula. This is a workforce challenge as well as a financial problem since many of these institutions serve areas where the relevant talent does not exist and cannot competitively be recruited and then retained given the highly sought-after skills required, nor is it financially sensible to maintain such staff at most of these colleges, as their size and budgets do not warrant it.

The NRP Socio-technical Solution

We developed technologies and processes that allow a small team of system administrators and user support personnel to manage a national-scale infrastructure across thousands of colleges. It is well known, and well-practiced by commercial cloud providers, that the human effort to operate compute and data infrastructure scales much less than linear, likely logarithmically, with the amount of equipment under management. We added to this the ability to manage equipment irrespective of location. A college with an existing Science DMZ [3] thus may purchase equipment from a broad list we recommend, rack it up, connect power and networking, and we take over management of the system from there. As a result, we eliminate the critical recurring operational and security support at each college, by centralizing and automating operations, security, and user support in a scalable fashion (by critically maintaining all nodes at the same software release levels, and constant monitoring for improper usage, failed nodes, etc.). This not only reduces the total cost of ownership, but also improves cybersecurity by bringing all colleges to the standard of the San Diego Supercomputer Center (SDSC), a national scale facility at UC San Diego, and the Holland Computing Center at the University of Nebraska-Lincoln by dint of us operating the entire national-scale NRP cluster.

In addition, we have already adapted social network and AI techniques to provide scalable user support to the community of educators on the NRP platform. We built a community of educators and researchers that actively share best practices in chat channels on a wide range of topics, supervised by our expert personnel. To further scale the support, we use the chat transcripts in our chat channels to train AI chat bots to answer questions from the user community.

At present, our prototype includes hardware at 45 colleges, and 8 Research and Education Networks (RENs) across the USA and Internet2 [4]. Internet2 is the national network provider that connects the RENs with each other. We have a strategy to scale this out to a thousand colleges to provide a million students per year with AI Education Infrastructure as described below.

The NRP Strategy to Scale out to the nation

Our strategy is to work with RENs like CENIC [5] and the Great Plains Network (GPN) [6]. CENIC is a 501(c)(3) with the mission to advance education and research across the state of California by providing the world-class network essential for innovation, collaboration, and economic growth. Its charter members include the California K-12 system, the California Community Colleges (CCC), the California State University System (CSU), the University of

California System (UC), Stanford, Caltech, USC, and the Naval Postgraduate School, and most of the California Public Libraries. The 116-campus CCC system alone has more than 2 million students enrolled, providing education to more than 10% of all college students in the USA [7]. The GPN is a peer organization to CENIC that serves many Midwest states: South Dakota, Nebraska, Kansas, Missouri, Oklahoma, Arkansas, as well as affiliated members in Wisconsin, Iowa, and Minnesota. All the RENs in the USA coordinate with each other via the Quilt [8], a national coalition of non-profit US regional research and education networks, and are all linked with each other, and the general global internet by Internet2.

As a collective and in addition to their technical role, the RENs of the USA form a “social” network that connects a vast number of institutions of higher learning, both public and private, encouraging collaborations via fiber, packets, and people. Based on our existing work with 8 RENs, including CENIC and GPN, and Internet2, as well as attendance at the annual Quilt meetings over many years, we know that most RENs are very keen to offer AI compute services to their academic constituencies but, not being their core focus, lack the staff expertise to do so.

A $25M investment over 5 years, followed by a steady state operations costs of $8M/year will allow us to achieve our goal of a thousand colleges and a million students served per year. The infrastructure operations costs would thus amount to only $8 per student per year, an amount negligible compared to the administrative costs of each college maintaining its own infrastructure including the costs to support and train the educators on such infrastructure. During the first 5 years, we would scale up our workforce from 3.5 FTE today to 24 FTE needed for 24x7 operations of NRP across our thousand colleges, and spend ~$10M for hardware that we would install at strategically selected institutions across the USA. We have found that initial hardware seeding in campus Science DMZs is exceptionally useful to stimulate future investments, and generate excitement, especially at smaller institutions. We propose that the hardware funds would be exclusively earmarked for non-R1 institutions and institutions in EPSCoR states. The $10M in hardware purchases will allow us to install hardware in ~150 institutions, motivating the other 850 institutions to make their own hardware investments. Over time, we expect all institutions to augment their hardware to meet their student needs.

After the initial 5 years, we would expect the NRP for AI Education to transition to a long-term sustainable model of operations. Internet2 and many of the RENs operate as 501(c)(3) nonprofits and fund their operations mostly via membership fees. This business model seems natural to transition NRP to. A combination of some of the RENs and Internet2 might offer NRP as a community cloud service to their communities, providing a complement to commercial cloud services. We thus propose that the Federal Government with its AI initiative fund a necessary transformative change for college education nationwide that can be sustained without ongoing Federal funding.

While this transformative change is necessary for college education, it also expands the national research capacity to a much wider range of academic institutions. This in turn has positive effects on local economies, especially in rural areas. The colleges we reach with NRP AIRE tend to have faculty engaged with the needs of their local economies on workforce development and applied research relevant to those local economies. By encountering research methodologies in college the workforce is much better prepared for the kind of continuous improvement processes often required from them in industry.

[1] https://link.springer.com/chapter/10.1007/978-94-009-2699-8_8, for example
[2] https://aisurvey.sdsu.edu/dashboard/
[3] https://fasterdata.es.net/science-dmz/
[4] https://internet2.edu
[5] https://cenic.org
[6] https://www.greatplains.net
[7] https://www.cccco.edu
[8] https://www.thequilt.net/